

Datasheet

Black Oak Engineering

precision electronic instruments

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LittleBear – Temperature Logging Verification System



Part number BE-LBTV
Version 1.0



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Description. **LittleBear** is a compact, battery-powered, NIST-traceable, laboratory temperature measurement device with 0.1 °C accuracy over an unusually broad range: -55 to 85 °C. It has enough battery capacity and memory storage to log data for months. Data are time stamped. It uses a USB-C interface to extract data. BOE's free **EasyPlot** app may be used to archive data in spreadsheet or PDF form. This combination will satisfy the most rigorous standards of ISO 17025, GMP, GLP, and USP.

LittleBear is designed for use in the laboratory, in industry, and in the field. User settings are easily programmed. The case may easily be mounted to a surface. Its compact size lets it hide away discretely in a corner while it does its job.

Specifications

- Range: -55 to 85 °C (-67 to 185 °F).
- Accuracy: ± 0.1 °C, calibrated to NIST standard at factory.
- Battery capacity: 1 A-hr.
- USB-C connection for programming and data output.
- Memory: 32 kByte EEPROM (retains memory even if power lost).
- Certifications: CE, IEC for laboratory instrumentation, REACH, RoHS3.
- .Step model available from BOE.

Mechanical & Environmental

- Enclosure is black ABS with rough surface, UL94.
- Dimensions: 48 × 48 × 20 mm (1.9 × 1.9 × 0.76 inch) with two 8 mm (0.3 inch) side flanges.
- Mass: 37 g (1.3 oz).
- Humidity / water exposure: IP54. Mount with USB access on bottom. Dust protected and water splash tolerant.
- Allow **LittleBear** to reach normal working temperature before extracting data. 20 minutes is generally sufficient.

Approvals & Compliance

- RoHS. REACH.
- California Prop 65. There are no listed toxic materials in the PCBA.
- CE. The system is designed and tested to meet or exceed CE requirements for laboratory, non-safety-critical electronic devices.

BOE is continuously improving. We also strive to keep one step ahead of procurement shortfalls. We will deliver to you the latest hardware version possible. In some cases specifications will change.

Warranty Policy. Any instrument ordered from BOE may be returned for full refund, less shipping costs, within 30 days of delivery, provided that the instrument has not, in the opinion of BOE been damaged or misused. An RMA number is required in all cases. See our *Standard Terms & Conditions – Instruments* for more details.

BOE reserves the right to make changes to these specifications as it deems necessary. All technical information contained herein is as accurate as possible; however BOE shall not be held responsible for any errors or for product use, nor for any infringements upon the rights of others which may result from its use. BOE products are not to be used in life support or safety critical applications.

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